

PHYSICS-INFORMED BUT NOT PHYSICS-CONSISTENT: ERROR GEOMETRY AND SUBSPACE PROJECTION FOR NEURAL AC POWER FLOW

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ABSTRACT

Recent neural power-flow solvers, including emerging foundation models, achieve accurate voltage predictions, yet such accuracy does not necessarily imply physically consistent solutions. Even small complex voltage errors can yield large AC power-balance residuals. We study this accuracy–consistency gap across PIGNN-GC, GridSFM, gridfm-graphkit, and LUMINA on realistic 2224-bus Great Britain network (GBnetwork) scenarios, with cross-grid evaluation of GridSFM over 31 systems. Using a singular value decomposition (SVD) basis fitted to training AC power-flow solutions, we find that neural prediction errors contain substantial components outside the dominant solution subspace. To address this mismatch, calibrated solution-subspace projection (CSP) suppresses off-subspace prediction components after train-only bias calibration, reducing Mean PB by 67.0%, 37.8%, 40.5%, and 68.9% for PIGNN-GC, GridSFM, gridfm-graphkit, and LUMINA, respectively, relative to calibrated predictions, while improving voltage-magnitude accuracy in all four models. These results identify output-error geometry as an important factor in physics-consistent neural AC power flow. Code: github.com/Kimchangheon/neural-acpf-error-geometry

Index Terms— AC power flow, physics-informed neural network, graph neural network, error geometry, subspace projection

1. INTRODUCTION

AC power flow (AC-PF) is a fundamental tool for steady-state power-system analysis [1]. It is repeatedly solved in congestion management [2] and contingency studies [3], and underlies optimal power flow (OPF) [4, 5] and state-estimation workflows [6]. The need for both computational efficiency and physical consistency has motivated neural surrogates including direct data-driven approximation [7, 8], physics-informed training [9, 10, 11], feasibility-seeking mechanisms [12, 13], and learned projection-based correction [14].

More recent cross-grid and foundation-model efforts—including GridSFM [15], gridfm-graphkit and GENCO [16, 17], and LUMINA [18]—extend this direction with global context and topology-aware representations of heterogeneous grid components [19] designed to operate across diverse grids and operating conditions [20].

However, accurate voltage prediction does not necessarily imply physical consistency [21, 22]. Physics-informed objectives can

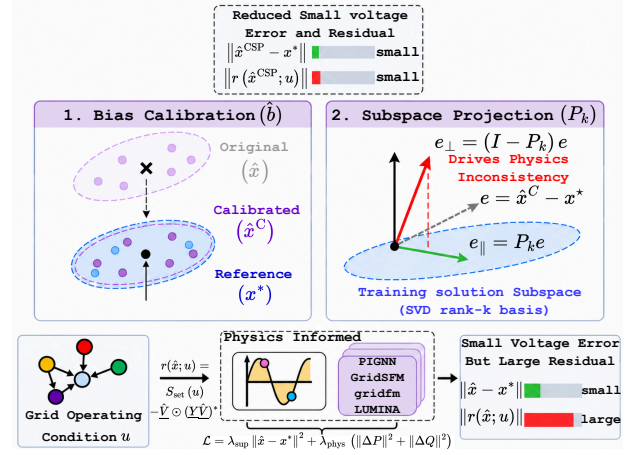


Fig. 1. A low-dimensional subspace is derived from the training solutions. CSP first removes train-estimated bias and projects the corrected prediction onto this subspace, suppressing off-subspace errors and improving voltage accuracy and AC power-balance consistency.

reduce power-flow residuals [23, 24], but remain soft training objectives and do not by themselves enforce exact satisfaction of the nonlinear AC equations [25]. Consequently, neural surrogates with small voltage errors can still exhibit non-negligible power-balance violations. We refer to this discrepancy as the *accuracy–consistency gap* and ask which properties of the remaining prediction error govern its physical impact.

We hypothesize that error *direction*, rather than magnitude alone, is an important factor: similarly sized voltage errors can induce substantially different AC residuals depending on their orientation across buses. Recent work has examined directional effects of warm-start errors in the distinct context of Newton–Raphson (NR) convergence [26]. This motivates studying the *geometry* of neural-surrogate errors in the voltage-state space.

Goodwin et al. [27] characterized the differential geometry of the AC-PF solution manifold, while Park et al. [28] and Chevalier et al. [29] demonstrated low-dimensional structure in AC-OPF and PF solutions, respectively. Motivated by this evidence of low-dimensional solution structure, we ask whether the relationship between neural prediction errors and a *training solution subspace* can be quantified to explain physical inconsistency, and whether this structure can be exploited to reduce power-balance violations.

To address these questions, we make three contributions. (i) We decompose calibrated prediction errors of four neural surrogates into

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in- and off-subspace components using a low-rank singular value decomposition (SVD) basis fitted to training AC-PF solutions, and show through fixed-norm interventions and local Jacobian analysis that off-subspace errors are substantially more residual-sensitive. (ii) Building on this finding, we propose *calibrated solution-subspace projection* (CSP), a lightweight post-hoc output correction that uses only training data, removes systematic bias, and suppresses off-subspace prediction components without retraining. (iii) We evaluate CSP on frozen PIGNN-GC, GridSFM, gridfm-graphkit, and LUMINA models on the 2224-bus GBnetwork, where it lowers AC power-balance residuals and voltage-magnitude error in all four models and provides a better initialization for one Newton update. We further assess its generality on 30 additional systems and its robustness under topology and load-distribution shifts.

2. ERROR GEOMETRY AND CSP

2.1. AC power-balance consistency and calibration

For an N -bus grid, let $x = [V, \theta] \in \mathbb{R}^{2N}$ denote the voltage state, with $V = V \odot e^{j\theta}$. Given operating condition u , a surrogate predicts $\hat{x} = \hat{f}_\phi(u)$, from which (P, Q) are recomputed using the network admittance model. For setpoints $(P^{\text{set}}, Q^{\text{set}})$, we define

$$r(\hat{x}; u) = \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} P^{\text{set}} - P \\ Q^{\text{set}} - Q \end{bmatrix}, \quad \text{PB} = \sqrt{\Delta P^2 + \Delta Q^2}.$$

PB uses ΔP at PV/PQ and ΔQ at PQ buses, zero elsewhere; prescribed PV/slack magnitudes and the slack angle are restored before scoring. Mean/Max PB aggregate all bus-scenario pairs on a common 100-MVA per unit (p.u.) base.

Before geometric analysis, we remove train-only prediction bias:

$$\hat{b} = \mathbb{E}_{s \in \text{tr}} [\hat{x}_s - x_s^*], \quad \hat{x}^C = \hat{x} - \hat{b},$$

where s indexes scenarios, $\star$ denotes the converged NR reference, and C denotes calibration. $\hat{b}$ is frozen at test time; angle bias is estimated as the per-bus circular mean of wrapped prediction errors.

2.2. Error geometry and training solution subspace projection

From training Newton-Raphson solutions, let $X_V, X_\theta \in \mathbb{R}^{S_{\text{tr}} \times N}$ stack centered magnitudes and circularly centered angles:

$$X_V = W_V \Sigma_V U_V^\top, \quad X_\theta = W_\theta \Sigma_\theta U_\theta^\top, \quad P_k = U_k U_k^\top, \quad (1)$$

where $U_k = \text{diag}(U_{V,k}, U_{\theta,k})$ contains the leading k state-space directions per channel; $k = 16$ denotes 16 per channel.

For calibrated error $e = \hat{x}^C - x^*$,

$$e^\parallel = P_k e, \quad e^\perp = (I - P_k) e, \quad \hat{x} - x^* = \hat{b} + e^\parallel + e^\perp. \quad (2)$$

Here, $\hat{b}$ is the train-estimated mean bias, while $e^\parallel$ and $e^\perp$ partition the remaining error into in- and off-subspace components.

Around a converged state with fixed topology and bus types,

$$r(x^* + e; u) = r(x^*; u) + J(x^*; u) e + \mathcal{O}(\|e\|^2), \quad (3)$$

where $r(x^*; u) = 0$ and J is the AC residual Jacobian; hence equal-norm errors can yield different residuals by orientation.

Motivated by this orientational sensitivity, CSP first removes $\hat{b}$ and then projects onto the dominant training-solution directions:

$$\boxed{\hat{x}^{\text{CSP}} = \bar{x}_{\text{tr}} + P_k (\hat{x} - \hat{b} - \bar{x}_{\text{tr}})} \quad (4)$$

where $\bar{x}_{\text{tr}}$ is the training-solution mean; both $\bar{x}_{\text{tr}}$ and P_k remain fixed at test time. Projection can remove prediction error but also discard valid reference variation. Let $e^C = \hat{x}^C - x^*$ and $e^{\text{CSP}} = \hat{x}^{\text{CSP}} - x^*$. Since P_k is orthogonal,

$$\|e^{\text{CSP}}\|_2^2 - \|e^C\|_2^2 = \|(I - P_k)(x^* - \bar{x}_{\text{tr}})\|_2^2 - \|(I - P_k)e^C\|_2^2. \quad (5)$$

Hence, CSP improves state accuracy when the removed off-subspace prediction-error energy exceeds the discarded reference variation.

CSP is a lightweight, data-driven post-hoc output correction rather than a projection onto the nonlinear AC-feasible set; it therefore does not guarantee exact feasibility, but can improve physical consistency when the removed off-subspace components are residual-sensitive under the local AC Jacobian.

3. EXPERIMENTS AND RESULTS

Our primary study uses 37,022 converged pandapower Newton-Raphson operating points on the 2224-bus GBnetwork, split 1:1:1 for training/validation/test. The dataset combines global load/generation scaling with independent load P/Q and active-generation perturbations; the backbone topology and PV-voltage control schedule remain fixed. We evaluate PIGNN-GC, our global-context extension of PIGNN-Attn-LS [25] retaining its line-search, GridSFM [15], gridfm-graphkit [16], and LUMINA [18]; GridSFM is also evaluated cross-grid on 31 systems.

All models are trained from scratch using model-specific stable initialization, and CSP is evaluated within each frozen model. gridfm-graphkit uses its released architecture/configuration without a pretrained checkpoint. Both GridSFM and LUMINA retain their released backbones and are adapted from AC-OPF to AC-PF by fixing generator dispatch, predicting only voltage states with PF-specific residuals, and applying known-variable masking. All models use pre-solution pandapower [1, 30] internal PPC-derived inputs. Training and inference use one NVIDIA H100 96-GB GPU; inference timing uses the largest memory-fitting batch to maximize GPU utilization. Calibration statistics and solution bases use training data only. From $k \in \{4, 8, 16, 32, 64\}$, validation Mean PB selects $k^* = 16$ without degrading calibrated voltage-magnitude RMSE and is frozen for test and distribution-shift experiments. Unless otherwise stated, mechanistic, ablation, and cross-grid analyses use model seed 42, whereas Tables 1 and 3 aggregate seeds 41–43.

Voltage RMSE alone can be small even when a model largely predicts a static per-bus profile rather than following scenario-dependent voltage variation [15]. To isolate this scenario-dependent response, we center each bus as $\tilde{V}_{is} = V_{is} - \bar{V}_i$ and $\tilde{V}_{is} = \hat{V}_{is} - \bar{V}_i$, and report

$$R_w^2 = \text{corr}^2(\tilde{V}_{is}, \tilde{\hat{V}}_{is}), \quad a_w = \frac{\text{Cov}(\tilde{V}_{is}, \tilde{\hat{V}}_{is})}{\text{Var}(\tilde{V}_{is})}. \quad (6)$$

R_w^2 is the squared correlation between the centered prediction and reference pooled over all bus-scenario pairs, while a_w measures the scale of the predicted variation relative to the true variation.

3.1. Off-subspace error and directional sensitivity.

Off-subspace error. Off-subspace error is substantial across models and is positively associated with Mean PB. Although the rank-16 basis explains 98.8%/99.7% of training voltage-magnitude/angle variation, the median fractions of total squared error lying outside the respective subspaces are 26.3/55.0/56.8/27.1% for magnitude

Table 1. GBnetwork results at $k = 16$. All models report mean $\pm$ standard deviation over three model seeds. C denotes calibration, P_{16} projection, and CSP_{16} their composition. Inference time is GPU-synchronized forward execution plus the corresponding output transform, reported per scenario.

Model	Output	V RMSE $\downarrow$	θ RMSE ($^\circ$) $\downarrow$	$R_w^2 \uparrow$	a_w	Mean PB $\downarrow$	Max PB $\downarrow$	Inference (ms/scen.)
Per-bus mean	Baseline	.02341	6.847	.000	.000	.12084	5.88	–
PIGNN-GC	Raw	.02850 $\pm$.00232	1.108 $\pm$.175	.671 $\pm$.050	.546 $\pm$.072	.31592 $\pm$.03240	195.89 $\pm$ 52.54	10.034 $\pm$.023
	C	.01393 $\pm$.00105	.777$\pm$.161	.671 $\pm$.050	.546 $\pm$.072	.16661 $\pm$.02352	190.00 $\pm$ 37.53	10.028 $\pm$.023
	P_{16}	.01624 $\pm$.00254	1.091 $\pm$.166	.829$\pm$.045	.541 $\pm$.071	.07594 $\pm$.00422	5.07 $\pm$.85	10.036 $\pm$.023
	CSP_{16}	.01222$\pm$.00134	.835 $\pm$.141	.829$\pm$.045	.541 $\pm$.071	.05497$\pm$.00257	3.93$\pm$.07	10.034 $\pm$.032
GridSFM	Raw	.02868 $\pm$.01652	2.294 $\pm$.302	.768 $\pm$.151	.724 $\pm$.149	.26867 $\pm$.13004	9.10 $\pm$ 4.02	25.045 $\pm$.207
	C	.01086 $\pm$.00403	1.781$\pm$.055	.768 $\pm$.151	.724 $\pm$.149	.08251 $\pm$.01093	7.29 $\pm$ 3.53	25.015 $\pm$.096
	P_{16}	.01814 $\pm$.01145	2.244 $\pm$.325	.936$\pm$.032	.724 $\pm$.149	.07630 $\pm$.01942	5.08 $\pm$.49	25.087 $\pm$.279
	CSP_{16}	.00782$\pm$.00307	1.807 $\pm$.053	.936$\pm$.032	.724 $\pm$.149	.05136$\pm$.00143	4.94$\pm$.42	24.577 $\pm$.091
gridfm-graphkit	Raw	.00892 $\pm$.00154	.385 $\pm$.042	.905 $\pm$.030	.811 $\pm$.045	.09282 $\pm$.01763	19.82 $\pm$ 8.65	5.917 $\pm$.019
	C	.00753 $\pm$.00118	.367$\pm$.042	.905 $\pm$.030	.811 $\pm$.045	.07685 $\pm$.01307	18.88 $\pm$ 8.69	5.730 $\pm$.004
	P_{16}	.00617 $\pm$.00099	.503 $\pm$.018	.959$\pm$.008	.801 $\pm$.044	.04593 $\pm$.00156	5.36 $\pm$.56	5.723 $\pm$.011
	CSP_{16}	.00606$\pm$.00090	.496 $\pm$.021	.959$\pm$.008	.801 $\pm$.044	.04571$\pm$.00135	5.08$\pm$.18	5.779 $\pm$.012
LUMINA	Raw	.02653 $\pm$.00031	2.703 $\pm$.313	.443 $\pm$.014	.346 $\pm$.012	1.29896 $\pm$.85056	664.19 $\pm$ 448.00	7.548 $\pm$ 0.369
	C	.01780 $\pm$.00020	1.442 $\pm$.060	.443 $\pm$.014	.346 $\pm$.012	.21432 $\pm$.01186	20.65 $\pm$ 15.82	7.380 $\pm$ 0.155
	P_{16}	.01898 $\pm$.00049	2.160 $\pm$.268	.906$\pm$.012	.338 $\pm$.012	.07641 $\pm$.00128	5.47 $\pm$.56	7.981 $\pm$ 0.868
	CSP_{16}	.01570$\pm$.00024	1.282$\pm$.058	.906$\pm$.012	.338 $\pm$.012	.06662$\pm$.00074	4.61$\pm$.19	7.891 $\pm$ 0.567

and 10.3/2.7/39.6/31.7% for angle, for PIGNN-GC/GridSFM/gridfm-graphkit/LUMINA, respectively. After accounting for differences in total error norms, scenarios with larger off-subspace fractions have higher Mean PB, with partial Spearman correlations of .659/.635/.594/.833 for magnitude and .789/.906/.809/.860 for angle (all 95% CIs exclude zero).

Fixed-norm intervention. At fixed error norm, suppressing off-subspace error consistently lowers Relative Mean PB across all four surrogates, whereas suppressing in-subspace error has the opposite effect (Fig. 2), isolating error orientation from error magnitude. We test this with an oracle diagnostic using the known reference state, varying the voltage-magnitude error orientation at fixed ℓ_2 norm:

$$\tilde{e}(\alpha) = \|e_V\|_2 \frac{e_V^\parallel + \alpha e_V^\perp}{\|e_V^\parallel + \alpha e_V^\perp\|_2}. \quad (7)$$

where phase angles remain at their calibrated predictions. Decreasing α suppresses off-subspace error; the control analogously suppresses the in-subspace component by exchanging $e_V^\parallel$ and $e_V^\perp$.

Jacobian sensitivity. Off-subspace directions are substantially more residual-sensitive than in-subspace directions under the local AC Jacobian, consistent with the fixed-norm intervention. Using directional gain $g(d) = \|Jd\|_2/\|d\|_2$, 32 random direction pairs at each of 128 held-out operating points yield a median paired off-/in-subspace gain ratio of 1,563 $\times$. The same gain contrast appears in the actual calibrated prediction errors: the ratios of median off- to in-subspace gains are 22.81 $\times$, 575.83 $\times$, 4.69 $\times$, and 6.51 $\times$ for PIGNN-GC, GridSFM, gridfm-graphkit, and LUMINA, respectively. The observed error regime is consistent with the first-order expansion in Eq. (3), with median linearized-to-nonlinear residual-change ratios between 0.996 and 1.009 across all four models.

3.2. Solution subspace specificity.

Our matched-control experiments show that preserving the same amount of reference-state variation is insufficient for low PB, indicating that the specific state-space directions retained by the solution

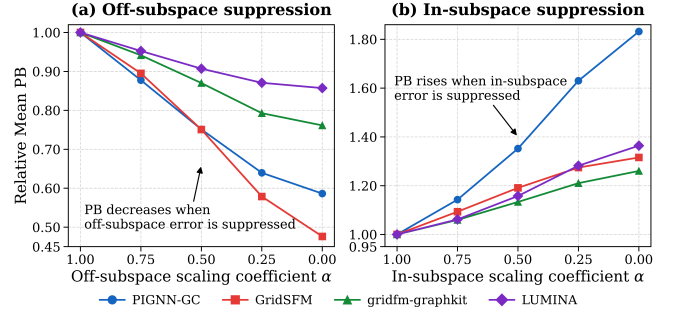


Fig. 2. Fixed-norm directional intervention on GBnetwork, with Relative Mean PB normalized at $\alpha = 1$.

Table 2. Basis controls on GBnetwork for PIGNN-GC. Parentheses denote V/θ ranks or smoothing strengths.

Control	Ref. var. retained (%)	Ref. PB	Pred. PB
SVD (16/16)	98.789/99.683	.0415	.0569
Laplacian matched (1750/2122)	98.819/99.688	26.15	26.22
Spatial matched ($\beta = .357/.208$)	98.801/99.684	6.20	6.22

subspace matter. We compare SVD₁₆ with graph-Laplacian and spatial low-pass controls, each tuned using training references to match the V/θ variation retained by SVD₁₆. On the test set, all three preserve nearly identical reference variation (Table 2), yet the controls yield substantially larger PB for both projected references and corrected predictions.

3.3. Post-training CSP correction

Table 1 compares Raw, C , P_{16} , and CSP_{16} across the four frozen surrogates. Relative to C , CSP reduces Mean PB by 67.0%, 37.8%,

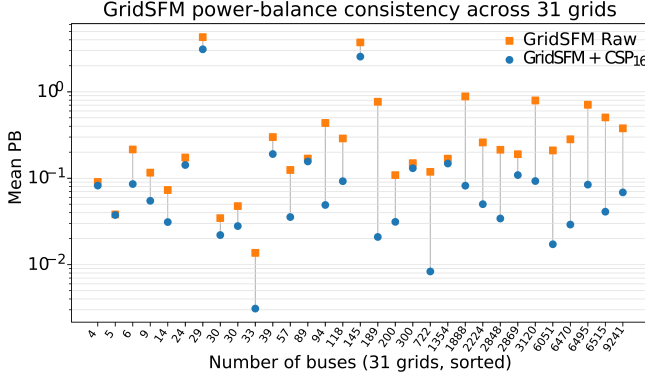


Fig. 3. Grid-by-grid physical consistency of GridSFM before and after CSP_{16} across 31 systems ordered by bus count. Mean PB is shown on a logarithmic scale.

40.5%, and 68.9%, respectively, while lowering voltage-magnitude RMSE in all four models. Projection improves R_w^2 with little change in a_w ; angle accuracy is model dependent. Raw and CSP_{16} inference times are comparable.

Magnitude-only, angle-only, and full projection show model-dependent effects of angle projection. PIGNN-GC achieves the lowest PB with full CSP, whereas GridSFM performs best with magnitude-only projection. For gridfm-graphkit, adding angle projection reduces Mean/Max PB from .0612/18.23 to .0457/5.08, but degrades angle RMSE from .367° to .496°. For LUMINA, it reduces Mean/Max PB from .1939/38.53 to .0660/4.51 while improving angle RMSE from 1.374° to 1.215°. Thus, better physical consistency from angle projection need not imply better angle accuracy.

These model-dependent accuracy changes are explained by Eq. (5). For magnitude, the removed off-subspace prediction-error energy exceeds the discarded reference variation across all four models. For angle, the balance differs across models: in gridfm-graphkit, angle projection discards more valid reference variation than model error and therefore worsens angle RMSE, whereas the opposite holds for LUMINA.

3.4. Complementarity with one-step Newton refinement.

CSP provides a consistently better initialization for one-step Newton refinement, improving both state accuracy and post-update physical consistency across all four models. Starting from C or CSP_{16} , we apply one Newton update (NR_1) to non-slack angles and PQ voltage magnitudes, with validation-selected damping factor; $\eta = 1$ is selected for all models, with no linear-solve failures. Relative to C initialization, CSP lowers magnitude and angle RMSE in all models and reduces post-update Mean PB by 21.1–96.4% (Table 3).

The benefit is particularly pronounced for LUMINA: $C + \text{NR}_1$ shows a large one-seed overshoot despite a successful linear solve, whereas CSP_{16} initialization remains consistent across seeds.

3.5. Cross-grid breadth and distribution shift

Cross-grid breadth. CSP reduces GridSFM Mean PB across all 31 systems (Fig. 3), spanning 4–9,241 buses and sub-kV to 750-kV levels. The set includes 28 pandapower/PYPOWER-format systems from IEEE, PEGASE, RTE, and GB benchmark families, plus three Common Information Model (CIM)-based networks. For each system, calibration and subspace statistics are fitted from its training data,

Table 3. One Newton update from calibrated or CSP_{16} outputs on GBnetwork. V RMSE and Mean PB are reported in 10^{-3} . Values are means $\pm$ standard deviations over three model seeds.

Model	Start	V RMSE (10^{-3})	θ RMSE ($^\circ$)	Mean PB (10^{-3})
PIGNN-GC	C	$1.80 \pm .12$	$.119 \pm .013$	$2.84 \pm .46$
	CSP_{16}	$1.17 \pm .10$	$.073 \pm .009$	$1.48 \pm .13$
GridSFM	C	11.86 ± 6.77	$.681 \pm .380$	2.50 ± 1.54
	CSP_{16}	$.97 \pm .20$	$.059 \pm .013$	$1.22 \pm .29$
gridfm-graphkit	C	$.943 \pm .232$	$.0843 \pm .0236$	$.941 \pm .325$
	CSP_{16}	$.496 \pm .085$	$.0277 \pm .0041$	$.742 \pm .080$
LUMINA	C	13.71 ± 9.23	$.671 \pm .038$	46.08 ± 64.15
	CSP_{16}	$1.040 \pm .016$	$.0802 \pm .0031$	$1.653 \pm .036$

Table 4. PIGNN-GC with fixed-rank CSP_{16} under topology and operating-distribution shifts. T transfers the control calibration and basis, whereas R refits both on condition-matched training data.

Condition	V RMSE	$R_w^2 \uparrow$	a_w	Mean	Med.	p95	p99	Max
Control	.01281	.8476	.4866	.0581	.0131	.2665	.5462	3.58
$N-1$ (T)	.01314	.8391	.4750	.0603	.0131	.2713	.5632	33.16
$N-2$ (T)	.01288	.8458	.4870	.0617	.0132	.2713	.5721	27.45
GBcorr (T)	.01966	.4819	.4025	.1003	.0231	.4512	.9746	7.00
GBcorr (R)	.01620	.5150	.4247	.0680	.0174	.3031	.6415	4.99

using nominal $k = 16$ per channel and the available SVD components on smaller systems. PV-voltage setpoints vary for most systems, but are fixed for GBnetwork with conflicting controllers.

Topology and spatial shifts. CSP remains stable in most statistics under $N-1/N-2$ topology shifts, with degradation concentrated in isolated worst-case violations, whereas GBcorr causes a broader performance drop. Under the topology shifts, voltage accuracy, tracking, and bulk PB remain close to control, while larger maxima motivate worst-case assessment beyond average error. GBcorr applies correlated, spatially varying load scaling across 14 zones while preserving total demand and generation settings. Transferring both calibration and basis gives Mean PB ≈ 0.100 p.u.; refitting calibration alone reduces this to 0.0944, basis alone to 0.0740, and both to ≈ 0.068 . Although the transferred/refitted bases retain 93.55%/98.53% of GBcorr magnitude variation, projecting converged GBcorr references yields Mean PBs of 0.0868/0.0438 p.u., respectively. This reference-only diagnostic isolates the effect of basis transfer from surrogate error and helps explain the improvement from basis refitting.

4. CONCLUSION

This work identifies output-error geometry as an important factor in the gap between voltage accuracy and physical consistency in neural AC power-flow surrogates. Across four architecturally diverse neural surrogates, we show that small voltage errors can contain off-subspace components that induce large AC power-balance violations. Building on this insight, calibrated solution-subspace projection suppresses these components, reducing voltage-magnitude error and power-balance residuals without retraining; the residual reduction extends to 30 additional grids. These results suggest that error direction, not only magnitude, should inform surrogate evaluation and design. They also motivate extending physics-consistent neural surrogates toward broader power-system analysis, estimation, and optimization.

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